A-A-54187A
23 November 1991
SUPERSEDING
A-A-54187
25 June 1990

COMMERCIAL ITEM DESCRIPTION

FREEZER, MECHANICAL, BLOOD PRODUCTS

(Ultra Low Temperature, Chest Type, 700 and 300 Unit Capacity)

This Commercial Item Description covers two types of ultra-low temperature biological storage freezers.

Salient Characteristics:

General characteristics. The freezers shall be chest style, ultra low (minus 121°F) temperature freezers suitable for freezing and maintaining biological products/blood products, including red blood cells, plasma, platelets, and cryoprecipitates. The freezers shall be furnished in the following types and capacities:

Type I - 700 unit capacity

Type II - 300 unit capacity

Chamber capacity.

Type I freezer shall have sufficient storage capacity to accommodate 700 units of blood product contained in standard blood boxes, stored flat side down. Filled and bulging dimensions of boxes are 7-1/8 inches by 5-5/8 inches by 1-7/8 inches.

Type II freezer shall have sufficient storage capacity to accommodate 300 units of blood product contained in standard blood boxes, of dimensions stated in above paragraph, and loaded in the manufacturer's standard type racks.

Construction material. The interior and exterior of the freezer shall be fabricated of stainless, or corrosion protected, steel plate having a minimum thickness of 22 gage.

AMSC N/A FSC 4110

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:30Z
Check the source to verify that this is the current version before use.

Insulation. The lid, walls, and bottom of the freezer shall be insulated with rigid urethane foam, poured or foamed-in-place under pressure in a heated foaming station. The insulation thickness shall be 3-1/2 inches in lid, 3 inches in side walls and bottom.

Tolerance of plus or minus 3/16 inches shall be maintained for insulation thickness. No wiring, sensors, or devices which will require maintenance shall be placed inside or be covered by the foam.

Refrigeration system(s). Type I freezer shall be equipped with dual cascaded refrigeration systems (4 motor/compressor combinations). Each motor/compressor combination shall be provided with an on/off switch. Each cascaded system shall contain a high stage and a low stage. The evaporator of each low stage refrigeration system shall contain its own copper tube circuit, which is fastened by welded stud, refrigerating all four sides of the chamber (and the bottom of the chamber if copper tube circuit is also used on the bottom). Each circuit of the refrigeration system, shall be designed to uniformly freeze the chamber; therefore, regardless of which cascade refrigeration system is operating, the chamber shall be uniformly refrigerated.

Type II shall be equipped with a single cascaded refrigeration system identical to the type I systems. It shall also be equipped with a back-up system of the type specified at time of order: either another cascaded system or a CO_2 cooling system.

Each freezer compressor shall be the reciprocating accessible semi-hermetic type; each compressor shall be a minimum of one horsepower. Each freezer compressor shall be provided with an oil level sight glass, to visually observe proper refrigerant level, and a brass service valve, to facilitate the use of gauges and the addition of refrigerants during servicing. The motors driving the compressors shall operate at 1750 rpm and be rated at least one horsepower each. The compressors shall use non-flammable refrigerants.

Cooling. The refrigeration system shall be configured to have capability of air cooling, water cooling, or a combination of both, as specified by the type number and customer order. Type I shall specify either air, water, or a combination of both air and water cooling. Type II shall specify either air or water cooling but not the combination of

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:30Z
Check the source to verify that this is the current version before use.

air-water cooling. The cooling systems' designs shall allow sufficient reserve capacity to maintain an internal chamber operating temperature of at least minus 121°F or colder with air cooling for ambient air temperature of 95°F plus or minus 2°F, or with water cooling for water temperature of 85°F plus or minus 2°F. For systems which use air cooling, the air inlets to the refrigeration condensers shall be equipped with removable filters, with capability to be cleaned with soap and water and reused continuously.

Door lid. The freezer shall be supplied with magnetically sealed door lid gasket(s) constructed with material which remains flexible and usable at the expected internal temperatures in use. The freezer shall have a minimum of three movable or removable sublids constructed of Lexan or equal. Sublids shall allow access to part of the contents without exposing the entire contents to ambient conditions when the freezer is opened. The freezer lid shall be counter balanced or strutted to hold any position from closed to fully opened with a maximum movement force of ten pounds in either direction.

Anti-icing. The unit shall be supplied with anti-icing to the extent that it will prevent freezer lock of the freezer door lids and of the contents of the freezer when the unit is operated in a 95°F plus or minus 2°F, 95 percent relative humidity plus or minus 5 percent at sea level atmospheric pressure. In these ambient conditions, opening the lid fully for 60 seconds plus or minus 2 seconds, every 4 hours for a 24 hour period, shall produce no frost build-up or freeze-lock of doors (outside lid or inner sliding doors) or contents with a full, half-full, or empty freezer. Formation of non-adhering ice crystals is acceptable.

Door locks. The freezers shall be provided with two (2) cam actuated locking mechanisms with one padlock and key for each cam lock. The two cam locks shall be located on the front of door lid and freezer, from 1/4 to 1/3 of the freezer lid length from each end of the lid. The lock mechanisms shall permit key removal in either locked or unlocked position.

Temperature indicating devices. A digital temperature indicator shall be provided to indicate freezer chamber temperature (for Type I). Indicated temperature shall be within plus or minus 5°F of actual temperature. The indicator shall be line powered and backup battery powered to enable continuous temperature indication even during power failure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:30Z
Check the source to verify that this is the current version before use.

A seven day continuous freezer chamber temperature recorder shall also be provided (for Type I). It shall be a single pen, inkless, pressure writing system with a seven day per revolution drive. The recording range shall include from 75°F through minus 153°F. The temperature accuracy shall be within plus or minus 5°F. It shall be easily visible and accessible for both temperature observation and for changing charts. The recorder shall record temperature even during power failure.

Alarm systems. The freezers shall be provided with line powered and backup battery powered audible and visual alarms which shall activate for power failures and temperature deviations beyond the pre-set warm and cold limits. Capability shall be provided for setting the warm alarm limit within 6°F to 9°F above the operating temperature set point; the cold alarm limit within minus 6°F to minus 9°F below the operating temperature, within the operating temperature range of at least minus 85°F through minus 121°F. The alarm set points shall be settable to within plus or minus 2°F. (For Type II, the alarm shall activate when the backup cooling system is activated, whether it is the CO_2 or cascaded refrigeration system.) The alarms shall have the capability of being silenced temporarily with a push button or similar device, or being defeated with a key switch. Contacts and compatibility shall be provided for connection of the alarm circuitry to a remote alarm or a computerized central alarm/monitor system. The alarm systems shall continue to function on battery power for a minimum of 5 hours when AC power is lost to the system. The alarms shall be provided with a test mechanism which shall allow testing alarm circuits for proper functioning.

Control panel. The freezer shall be provided with a control panel at the front. Included on this control panel shall be at least the following items:

- Chamber temperature digital indicator (for Type I).
- Chamber temperature chart recorder (for Type I).
- Chamber operating temperature setpoint mechanism(s).
- Warm and cold limit alarm setpoint mechanisms.
- Warm and cold limit alarms-audio and visual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:30Z
Check the source to verify that this is the current version before use.

- Power failure alarms-audio and visual.
- Audio temporary silence button or mechanism.
- Key operated alarm 'defeat' switch.
- Alarm test button or mechanism.
- Battery charge level indicator.
- Key operated refrigeration systems 'ON/OFF' switch.
- Backup system alarms-audio and visual (for Type II).
- Anti-icing heater switch.

Operating characteristics. The freezers shall be capable of maintaining operating temperatures at any operator selected temperature within the temperature range of at least minus 85°F through minus 121°F. A mechanism shall be provided for setting the operating temperature within 2°F of desired temperature with absolute accuracy of plus or minus 5°F. Operating temperature shall be controlled within plus or minus 6°F from readings at the sensing and control bulbs.

The compressors shall cycle at least once every 1/2 hour while internal temperature remains at minus 115°F or colder, and the operating temperature range shall not exceed plus or minus 6°F.

The freezer shall have pull down capability to cool an empty chamber from an initial temperature of 75°F to minus 121°F in 3 hours maximum without the aid of backup or any other additional cooling system.

While the internal temperature of the freezer is maintained at minus 121°F there shall be no increase in motor current demand when operated under the following conditions:

1. Ambient temperature plus 95°F plus or minus 2°F.
2. Cooling water temperature plus 85°F plus or minus 2°F.

For Type I, in normal operating mode to maintain internal temperature at a pre-set level, the cascaded systems shall cycle alternately from one system to the other in an ambient temperature of less than 90°F. For rapid cool down of the contents of the freezer, means shall be provided to allow

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:30Z
Check the source to verify that this is the current version before use.

automatic simultaneous operation of both cascaded refrigeration systems. Such operation shall be automatic when the internal temperature of the freezer exceeds the warm alarm set point. Failure of either cascaded refrigeration system shall cause the surviving cascaded system to function and maintain the pre-set operating temperature indefinitely. In addition, alarms visual and audible, shall signal the failure.

For Type II, cycling shall occur at ambients below 90°F. The freezer shall be equipped with the specified backup cooling system. This backup system shall automatically initiate cooling in the event of compressor system failure or power failure. Regardless of which type backup system is chosen, an audio and visual alarm shall be set off automatically to notify backup operation. A temporary or defeat mechanism shall enable alarm shut-off. The carbon dioxide (CO_2) backup system shall automatically maintain the internal temperature of the freezer chamber at minus 76°F for at least 48 hours. If a cascade refrigeration system is selected for backup, it shall be identical to the primary cascade system. The backup system shall automatically take over in the event of the primary system failure to maintain original performance of the freezer at minus 121°F indefinitely.

Casters. The freezers shall be provided with casters to facilitate easy movement.

Dimensions. Exterior dimensions of the freezer shall not exceed 110.0 cubic feet.

Labels. The freezers shall have a readily accessible permanently mounted label(s) with at least the following information:

- Item identification
- Manufacturer's name and Government assigned FSCM
- Contract number and date of manufacture
- Freezer's serial numbber
- National stock number (NSN)
- Electrical requirements
- Maximum sustained operating current draw

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:30Z
Check the source to verify that this is the current version before use.

- Locked compressor rotor current draw
- Low and high-side design pressures (in accordance with UL 471)
- Refrigerant type and amount required.
- Maximum current draw of battery charger
- Product intended for indoor or outdoor use

The above label information may be provided on two labels, if they are mounted adjacent to one another.

NOTE: Wiring system for connection shall not necessarily be listed on the same nameplate. The wiring marking shall be located so that it will be visible when power supply connections to the product are being made by the user.

Optional Equipment. When specified by the customer requisition, the following equipment shall be supplied with the freezer:

- Remote alarm indicators.
- Standby, automatically activated voltage booster to protect freezer in case of reduced voltage conditions (brown-out).
- For Type II when specified, sufficient standard type racks shall be provided, appropriately arranged to allow storage of at least 300 blood or plasma units of dimensions specified herein.
- For Type II when specified, temperature-indicator recorder/external temperature indicating device with a seven day continuous recording chart, single pen, inkless, pressure writing system shall be provided. Steady-state temperature variation shall not exceed plus or minus 5°F from the readings at the sensing and control bulbs. The temperature indicator shall run on battery power if the the line power fails.

Safety. The freezers shall conform to UL 471-Standard for Safety, Commercial Refrigerators and Freezers. Acceptable evidence of this shall be a UL listing mark or a certified test report from an independent laboratory acceptable to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:30Z
Check the source to verify that this is the current version before use.

Government.

Manuals. The contractor shall furnish manuals, handbooks and/or brochures containing complete operation, installation and service/maintenance instructions (including pictures of illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures will be written in the English language and all schematics and wiring diagrams will use American electrical and electronics symbols. The manual will include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name and quantity required. All service notes, service memos, etc., pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included. These instructions shall be, as a minimum, that furnished to service engineers (or serviceman) that normally install and service the equipment for the company or distributor. When the system being procured includes, or will operate ancillary equipment, the service data shall include complete instructions and drawings which show interfacing of all system components.

Two complete sets of the above information shall be packaged with each unit of the required equipment. Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, then only a total of two service manuals need to be supplied to each consignee. However, two operational manuals still need to be supplied with each unit.

Finish. The freezer shall be fabricated from materials which are inherently resistant to corrosion or which are treated to render them corrosion resistant. Where painting or plating is used, appropriate cleaning and primimg shall be performed prior to finishing. Finish shall provide a pleasing professional appearance and be free of gaps, scratches, chips and other imperfections.

Workmanship. The ultra low temperature, biological products/ blood products freezer shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One ultra low temperature, biological products/blood products freezer, of the type specified, constitutes one unit.

Electrical Characteristics:

The freezers' electrical characteristics shall be in accordance

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:30Z
Check the source to verify that this is the current version before use.

with the Type number and customer order.

Type I. The freezer shall operate over the full tolerance range of one or more of the following line voltage and frequency combinations as specified by the customer requisition.

220 VAC plus or minus 10 percent, 60 Hz plus or minus 2 Hz, three phase.

208 VAC plus or minus 10 percent, 60 Hz plus or minus 2 Hz, three phase.

Type II. The Type II freezer shall operate over the full tolerance range of the line voltage and frequency combinations as specified by the solicitation. Use of commercial voltage (220 V in lieu of 230 V) is acceptable.

230 VAC plus or minus 10 percent, 50 Hz plus or minus 2 Hz, single phase.

208 VAC plus or minus 10 percent, 60 Hz plus or minus 2 Hz, single phase.

230 VAC plus or minus 10 percent, 50 Hz plus or minus 2 Hz, three phase.

208 VAC plus or minus 10 percent, 60 Hz plus or minus 2 Hz, three phase.

Change-over devices, switches, circuits, or transformers if used for the purpose of voltage or frequency conversion, shall be clearly marked and identified in the appropriate documentation which accompanies the unit. Mechanical or electrical adjustments to effect the conversion shall not require the use of specialized tools or skills.

QUALITY ASSURANCE PROVISIONS:

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:30Z
Check the source to verify that this is the current version before use.

requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 with an AQL of 1 (percent defective) and an inspection level of II.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 1 (percent defective) and an inspection level of II.

The unit of product for sampling purposes shall be one freezer unless the requisition calls for quantities of different frequencies and/or voltage phases. In this case, each different frequency and voltage phase shall be considered a different lot and inspected accordingly.

Examination/inspection. The following examination/inspection shall be conducted to determine compliance with this specification, and with UL 471 Standard for Safety, Commercial Refrigerators and Freezers.

Cabinet. Inspect the cabinet and parts for any damage or missing parts.

Insulation. Inspect and examine the insulation: five inches of foamed-in-place polyurethane on each side and bottom, with 3 1/2 inches in the lid.

Capacity. Type I 700 units; type II 300 units in standard racks.

Dimensions. Overall external 110 cubic feet.

Marking. The marking shall be in compliance with this

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:30Z
Check the source to verify that this is the current version before use.

specification.

Cautionary marking. A cautionary marking shall be prefixed by "Caution/Warning" or Danger" in letters no less than 1/8 inch high. The markings shall be permanent and shall be located to be visible before or immediately upon removal of a cover, panel, or the like, which normally encloses or protects a hazardous moving part, hot part, or electrically live part. The markings shall not be on the back of a removable cover or panel.

Single line wire diagram. The wiring system, shall be located so that it will be visible when power supply connection to the product are made by the end user.

Factory test pressure. The manufacturer is responsible for submitting the evidence of compliance with American Society of Mechanical Engineers', (ASME), Boiler and Pressure Code, Section VIII/Pressure-Relief Valves Refrigerant Containing Parts Subjected to Corrosion Resistant Materiel.

Tests. Tests shall be conducted to determine compliance with this specification. Where feasible the same sample shall be used for the determination of two or more test characteristics. Tests shall include, but shall not be limited to, the following:

The freezer shall be tested for compliance with performance requirements as specified under the Salient Characteristics paragraphs. The operation of the anti-icing capabilities (three 24 hour tests, each at a different load capacity) shall be tested prior to testing of reserve capacity. Unless otherwise specified timing of testing shall commence when environmental conditions have been stabilized for at least one hour with the unit in operation.

1. The freezer shall be stored in a room maintained at 75°F with its lid open for a minimum of 24 hours in order to stabilize the chamber temperature. Temperature sensor probes shall be placed at the top, bottom and center of the front and back panels (1 inch in from left and right sides). Probes shall also be placed in center of all four panels and the bottom. The temperature and current draw shall be recorded during pull-down and for a minimum of 4 hours afterward. The freezer indicator temperature shall not differ by more than 5°F from the add-on probes during the entire test period. Current draw shall not exceed the manufacturer's published data. The freezer shall reach minus 121°F within three hours maximum. Battery charger current draw data shall also be recorded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:30Z
Check the source to verify that this is the current version before use.

2. While ambient temperature is between 70°F and 80°F, and freezer temperature stabilized at minus 121°F, the empty freezer shall cycle at least once every 1/2 hour during a 4 hour period and temperature excursions shall not exceed 11°F.

NOTE: Compressor cycling shall occur at ambient temperatures of less than 90°F, while maintaining an internal temperature of minus 121°F.

3. With ambient temperature stabilized for 24 hours at 95°F plus or minus 2°F, relative humidity between 90 and 95 percent, empty freezer, there shall be no lid lock caused by frost build-up for the 24 hour test period and the temperature shall be minus 116°F or colder. During this test period, the freezer lid shall be opened every 4 hours for 60 seconds. Temperature may be warmer than minus 82°F for three minutes after closing. This test shall be performed 3 times, once each for freezer empty, half full, and filled to capacity. DoD approved frozen red blood cell freezing containers filled with a fluid that mimics the heat transfer and capacity of red blood cell solutions shall be used. (The vendor shall supply these test materials.)

4. With conditions stabilized, ambient at 95°F plus or minus 2°F, cooling water temperature plus 85°F plus or minus 2°F, relative humidity at 90 to 95 percent, and full freezer chamber at minus 121°F, the unit shall maintain an internal temperature of at least minus 121°F for a period of 96 hours. Current draw shall not increase during the 96 hours.

5. The ambient air temperature shall be maintained at a minimum of 95°F. The full freezer shall be stabilized at minus 121°F. The line power to the freezer shall be shut off for a minimum of 24 hours, during which time the freezer lid shall be opened for 10 seconds every 3 hours. The CO_2 backup system shall maintain the temperature of the freezer colder than minus 76°F for the entire test period. (The temperature shall be recorded at least every 15 minutes.)

Where a cascaded refrigeration system is used for backup, the freezer temperature shall be maintained at minus 121°F during the entire test period.

6. The units shall be tested for remote, local, and audio/visual alarm systems which alert the operator of any abnormal temperature in the freezer or, if power is disconnected. The alarm systems shall be activated when temperatures exceed the warm or cold limit set points.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:30Z
Check the source to verify that this is the current version before use.

7. The temperature recorder provided with the freezer shall be tested by recording the temperature of the freezer during the above 96 hour and 24 hour tests (Performance tests 4 and 5).

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:30Z
Check the source to verify that this is the current version before use.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document.

4110-01-320-5621 (Type I)	FREEZER, MECHANICAL, BIOLOGICAL/ BLOOD PRODUCTS, Ultra Low Temperature, Chest Type, 700 Unit Capacity, 208-230 volt, 50-60 Hz, Three Phase, AC
4110-01-320-8801 (Type II)	FREEZER, MECHANICAL, BIOLOGICAL// BLOOD PRODUCTS, Ultra Low Temperature, Chest Type, 300 Unit Capacity, 208-230 volt, 50-60 Hz, Single/Three Phase, AC

MILITARY INTEREST:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 4110-0443

Location: FRZRCID.1DB

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:30Z
Check the source to verify that this is the current version before use.